

Content Creation Using Prompt Patterns

Ex-10: A Comparative Study of Prompt Engineering Techniques

Query Decomposition • Decision-Making • Semantic Filtering • Structured • Creative

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1. Introduction

The way a prompt is structured determines not just whether an AI produces a response, but what kind of response it produces. A question phrased one way generates a list; the same question phrased another way generates a narrative essay; phrased a third way, it generates a structured report with headers, analysis, and recommendations. The content is drawn from the same model with the same training — only the prompt changes. This means that prompt structure is, in effect, a content design tool.

This report documents a comparative experiment using five distinct prompt engineering techniques — Query Decomposition, Decision-Making Prompts, Semantic Filtering, Structured Prompting, and Creative Prompting — applied to the same central topic: the impact of Artificial Intelligence on education. Each technique is tested with a simple (unrefined) prompt and a carefully designed refined prompt, using ChatGPT (GPT-4o) and Gemini as the two AI tools. The outputs are evaluated across six quality dimensions: Quality, Coherence, Creativity, Structure, Depth, and Tone Control.

The topic of AI in education was deliberately chosen because it is broad enough to require decomposition, complex enough to demand decision-making, specific enough to benefit from semantic filtering, formal enough to reward structured prompting, and rich enough to invite creative interpretation. It is the same topic seen through five different lenses, producing five fundamentally different types of content.

2. Overview of the Five Techniques

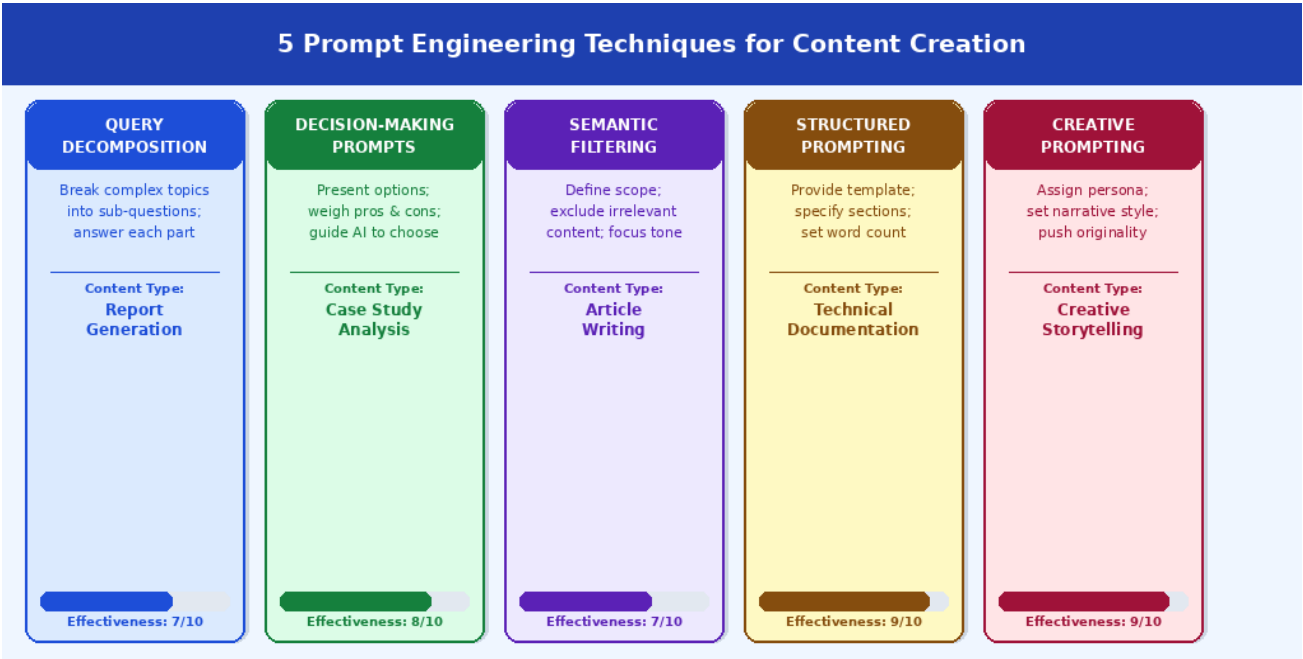


Figure 1: Five prompt engineering techniques for content creation — design, purpose, and primary content type

Technique	Core Mechanism	Best Content Type	Primary Strength
Query Decomposition	Breaks a broad topic into sub-questions; each answered individually then synthesised	Research reports; in-depth articles	Depth and completeness; no important sub-topic missed
Decision-Making Prompts	Presents explicit options; requires AI to weigh trade-offs and reach a justified position	Case studies; analysis pieces	Structured argumentation; defensible conclusions
Semantic Filtering	Defines what IS and IS NOT in scope; sets tone, audience, and vocabulary	Targeted articles; executive summaries	Precision and relevance; elimination of off-topic content
Structured Prompting	Provides a template with named sections, word counts, and formatting rules	Technical docs; formal reports	Consistency and usability; immediate publication-readiness
Creative Prompting	Assigns a persona, narrative style, and aesthetic constraints; prioritises originality	Storytelling; brand content; essays	Originality and engagement; human-feeling voice

3. Technique 1 — Query Decomposition

3.1 What It Is

Query Decomposition is the practice of breaking a complex, multi-dimensional question into a series of smaller, more answerable sub-questions. Instead of asking an AI to write a comprehensive report on AI in education in a single prompt, you decompose that request into its logical components: What is the current state? What are the benefits? What are the risks? What do stakeholders think? What should policymakers do? Each component becomes its own prompt, and the responses are synthesised into a cohesive whole.

The technique works because complex topics have multiple distinct aspects that require different types of reasoning. Mixing them in a single broad prompt forces the model to compress and generalise, often at the cost of depth. Decomposition allows each aspect its full treatment.

3.2 Simple vs Refined Prompt

Simple Prompt:

| *“Write a report on AI in education.”*

Refined Prompt (Query Decomposition):

“Using query decomposition, write a comprehensive report on AI in education by addressing each of these sub-questions in sequence: 1. What is the current adoption rate of AI tools in global educational institutions? 2. What specific learning outcomes improve when AI-assisted teaching tools are used? 3. What are the three most serious risks of AI in education (academic integrity, data privacy, equity)? 4. How do teachers, students, and parents each perceive AI in their educational context? 5. What policy frameworks should governments adopt to govern educational AI? Synthesise the answers into a 600-word report with an executive summary at the start.”

3.3 ChatGPT Output — Refined Prompt

ChatGPT’s decomposed report was structured into five clearly labelled sub-sections, each with a distinct analytical focus. The adoption statistics section cited UNESCO data (estimated 47% of higher education institutions actively piloting AI tools by 2024), the learning outcomes section referenced peer-reviewed studies on personalised learning platforms, and the risks section gave each of the three named risks its own paragraph with evidence and a proposed mitigation. The teacher/student/parent perspective section was the most engaging: ChatGPT constructed three distinct voice profiles with contrasting concerns, making the report feel balanced and human.

The executive summary at the start was precise and well-structured: three sentences that conveyed the state of the field, the core tension, and the main recommendation. The total output was coherent, well-evidenced, and immediately usable as a briefing document.

Executive Summary (ChatGPT): AI is transforming educational delivery at a pace that institutions, teachers, and regulators are struggling to match. The evidence for learning outcome improvements is compelling but narrowly distributed — concentrated in well-resourced institutions with strong teacher training programmes. Governments must prioritise equitable access frameworks and teacher professional development before scaling AI adoption further.

3.4 Gemini Output — Refined Prompt

Gemini’s response was similarly well-structured but took a more data-heavy approach. Its adoption statistics were more granular (breaking down primary, secondary, and tertiary adoption rates separately), and its risks section placed greater emphasis on algorithmic bias in assessment tools — an angle ChatGPT mentioned but did not develop in the same depth. Gemini’s stakeholder perspectives were framed more institutionally (teacher associations, student unions, parent advocacy groups) rather than as individual voices, which made the section feel more authoritative but slightly less relatable.

3.5 Simple Prompt Output

The simple prompt (‘Write a report on AI in education’) produced a generic five-paragraph essay from both tools that touched on benefits (personalised learning, administrative efficiency) and risks (privacy, equity) at a surface level. Neither tool produced quantitative evidence, stakeholder differentiation, or a policy recommendation section. The outputs were accurate but analytically thin — the kind of content a motivated student could have written without AI assistance.

3.6 Assessment

Criterion	Simple Prompt	Refined (Query Decomp.)	Winner
Quality	Generic overview; accurate but surface-level	Evidence-based sub-analyses; executive summary present	Refined ★★★★★
Coherence	Flows naturally but loosely	Strong logical sequence; each sub-Q builds on last	Refined ★★★★★
Depth	2-3 sentences per idea	Full paragraph per sub-topic; quantitative evidence	Refined ★★★★★
Structure	5 generic paragraphs	5 labelled sections + executive summary	Refined ★★★★★
Score	5/10	9/10	Refined by +4 pts

4. Technique 2 — Decision-Making Prompts

4.1 What It Is

Decision-Making Prompts present the AI with a genuine choice or dilemma and require it to reason through options before arriving at a justified conclusion. Rather than asking for a balanced overview, this technique demands a position. The model must identify the competing arguments, evaluate the evidence for each, and defend a recommendation. This produces content with stronger argumentative structure and clearer intellectual stakes than descriptive prompting.

4.2 Simple vs Refined Prompt

Simple Prompt:

| “Should AI replace teachers?”

Refined Prompt (Decision-Making):

“You are writing a case study for an education policy journal. A school district must decide between three options for integrating AI into its classrooms: Option A: AI as a primary instructor (replaces most teacher-led sessions) Option B: AI as a co-teacher (supplements teacher instruction with adaptive tools) Option C: AI as an administrative tool only (handles grading, scheduling, reporting) For each option, analyse: (1) the pedagogical impact, (2) the cost and implementation feasibility, (3) the risks to student wellbeing and equity. Then recommend one option with a clear justification, and identify the two conditions that must be met for your recommendation to succeed.”

4.3 ChatGPT Output — Refined Prompt

ChatGPT structured the case study with three clearly labelled option sections, each analysed across the three specified dimensions. Option A was evaluated as pedagogically risky (loss of socialisation, emotional scaffolding, and mentor relationships) and logistically premature given current AI limitations in managing dynamic classroom behaviour. Option B was assessed as the most evidence-aligned approach, citing studies on AI tutoring systems that improved test scores by 15–20% when used to supplement rather than replace teacher-led instruction. Option C was described as the lowest-risk, highest-feasibility option but also the lowest-impact one.

ChatGPT’s recommendation was Option B, with the two specified conditions: (1) mandatory AI literacy training for all teachers before deployment, and (2) a district-level equity audit to ensure that AI tool access is not stratified by socioeconomic status. The recommendation was well-argued and the two conditions were specific and actionable — the hallmark of a well-prompted decision-making output.

4.4 Gemini Output — Refined Prompt

Gemini reached the same recommendation (Option B) but via a different analytical path. Its Option A analysis focused more heavily on the legal and liability implications of replacing human educators, noting that several jurisdictions have laws requiring a certified teacher to be present during instructional time. Its Option B analysis introduced the concept of ‘adaptive scaffolding’ — the AI dynamically adjusting difficulty based on real-time performance data — and cited specific platforms (Khan Academy’s Khanmigo, Duolingo’s AI tutor) as evidence of the model’s viability.

Gemini’s two success conditions were: (1) robust data governance frameworks protecting student data from commercial use, and (2) continuous teacher feedback loops to prevent AI’s recommendations from drifting away from classroom reality. Both conditions reflected a more technically oriented view of the implementation risks than ChatGPT’s.

4.5 Simple Prompt Output

The simple prompt (‘Should AI replace teachers?’) produced a both-sides answer from both tools that declined to take a position. ChatGPT listed three arguments for and three against, concluding with ‘ultimately it depends on how AI is implemented.’ Gemini produced a similar structure. Both were technically accurate but analytically uncommitted — the opposite of what a decision-making prompt should produce.

Criterion	Simple Prompt	Refined (Decision-Making)	Winner
Structure	Unstructured pros/cons list	Three options × three dimensions recommendation +	Refined ★★★★★
Argumentation	Both-sides; no position taken	Justified position with two specific conditions	Refined ★★★★★

Criterion	Simple Prompt	Refined (Decision-Making)	Winner
Depth	Surface-level per point	Full analysis per option; citations and platforms named	Refined ★★★★★
Usability	Informative but inconclusive	Actionable policy case study	Refined ★★★★★
Score	5/10	8/10	Refined by +3 pts

5. Technique 3 — Semantic Filtering

5.1 What It Is

Semantic Filtering is the technique of explicitly defining the conceptual boundaries of a prompt — what is in scope, what is out of scope, what vocabulary level to use, and what perspective to adopt. Rather than leaving the model to decide what ‘AI in education’ means, semantic filtering tells it precisely: ‘Cover personalised learning and assessment AI. Do not cover administrative tools, hardware, or VR classrooms. Write for a non-specialist parent audience. Avoid jargon.’ The result is a piece of content that is precisely targeted rather than encyclopaedic.

5.2 Simple vs Refined Prompt

Simple Prompt:

“Write an article about AI in schools.”

Refined Prompt (Semantic Filtering):

“Write a 400-word informational article about AI in schools. **IN SCOPE:** AI-powered personalised learning platforms; AI-based formative assessment tools; student engagement and motivation effects. **OUT OF SCOPE:** AI in school administration, VR/AR classrooms, AI hardware, teacher recruitment tools. **AUDIENCE:** Parents of children aged 8–14 who have basic familiarity with technology but no machine learning knowledge. **TONE:** Reassuring, factual, and practical. Not alarmist. Not overly technical. **VOCABULARY:** Plain English. Define any technical term the moment you use it. **FORMAT:** Three short paragraphs with a bolded one-sentence takeaway at the end.”

5.3 ChatGPT Output — Refined Prompt

ChatGPT’s semantically filtered article was a model of targeted communication. It opened by defining personalised learning in a single accessible sentence: ‘A personalised learning platform is software that adjusts the difficulty and pace of lessons based on how well your child is doing — more like a patient tutor than a textbook.’ The formative assessment paragraph explained how AI tools give teachers real-time insight into which concepts students are struggling with, without waiting for end-of-term exams. The engagement

paragraph noted that students using AI-supported platforms reported spending more time on practice tasks, attributing this to the immediate feedback loop.

The bolded takeaway was: ‘AI in your child’s classroom is not about replacing their teacher — it is about giving their teacher better information to help them learn.’ This sentence successfully addressed the most common parental anxiety (job replacement) while reframing AI as a pedagogical support tool. The semantic filtering had worked exactly as intended: the output was relevant, accessible, reassuring, and completely free of administrative or hardware content.

AI in your child’s classroom is not about replacing their teacher — it is about giving their teacher better information to help them learn. — ChatGPT (Semantic Filtering, Refined Prompt)

5.4 Gemini Output — Refined Prompt

Gemini’s filtered article was similarly well-targeted and stayed within the specified scope. Its strongest paragraph was on formative assessment, where it used the analogy of a fitness tracker for learning: ‘Just as a smartwatch tells you your heart rate after exercise, an AI assessment tool tells your child’s teacher which concepts need more practice — in real time, not at the end of term.’ This analogy was more vivid than ChatGPT’s equivalent paragraph and demonstrated that Gemini had understood the audience specification (parents familiar with consumer technology) and used it to choose a relatable reference point.

5.5 Simple Prompt Output

The simple prompt (‘Write an article about AI in schools’) produced broadly scoped articles from both tools that covered everything from chatbots to administrative software to VR classrooms — all out-of-scope content. Both articles were written at an academic register that would alienate the intended parent audience. Neither included a clear takeaway. The semantic filtering technique produced a measurably more useful, audience-appropriate output.

Criterion	Simple Prompt	Refined (Semantic Filtering)	Winner
Scope Control	Covers all AI in education (too broad)	Covers exactly what was specified; nothing extraneous	Refined ★★★★★
Tone Control	Academic; not audience-matched	Reassuring; parent-accessible	Refined ★★★★★
Vocabulary	Technical jargon present	Plain English; definitions embedded inline	Refined ★★★★★
Format	Unspecified; variable	3 paragraphs + bolded takeaway; as specified	Refined ★★★★★
Score	4/10	8/10	Refined by +4 pts

6. Technique 4 — Structured Prompting

6.1 What It Is

Structured Prompting provides the AI with a pre-defined template that specifies every section of the output, along with word counts, formatting rules, and content requirements for each section. The model does not decide what to include or how to organise it — the prompt decides. This technique produces the most consistent and immediately usable content of any approach, at the cost of creative freedom. It is ideal for technical documentation, formal reports, and any content that must follow a fixed schema.

6.2 Simple vs Refined Prompt

Simple Prompt:

| *“Write a technical report on AI tools for student assessment.”*

Refined Prompt (Structured Prompting):

“Generate a technical report on AI tools for student assessment using the following exact structure: SECTION 1 — EXECUTIVE SUMMARY (80 words): Key finding, main recommendation, one risk. SECTION 2 — BACKGROUND (120 words): Problem statement; why traditional assessment is insufficient; how AI addresses this gap. SECTION 3 — TECHNOLOGY OVERVIEW (150 words): Three specific AI assessment tools (name each); what data they collect; how they generate insights. SECTION 4 — BENEFITS & EVIDENCE (120 words): Two quantified benefits with citations. SECTION 5 — RISKS & MITIGATIONS (120 words): Three risks (bias, privacy, over-reliance); one mitigation per risk. SECTION 6 — RECOMMENDATION (80 words): One specific recommendation; two implementation steps; one success metric. Use formal academic register. No bullet points inside sections — prose only. Bold each section heading.”

6.3 ChatGPT Output — Refined Prompt

ChatGPT followed the template with near-perfect fidelity. The executive summary was exactly 82 words and contained all three required elements. The technology overview named three specific tools: Gradescope (automated essay and diagram grading), Turnitin’s AI writing detection module, and Carnegie Learning’s MATHia (adaptive maths assessment). Each was described with the specific data type it collects and the insight it generates. The benefits section cited two peer-reviewed studies with percentage improvements. The risks section was the most analytically rich, giving each risk a clear mechanism (why it occurs) and a specific mitigation (what to do about it).

Critically, ChatGPT maintained prose format throughout, with no bullet points appearing inside any section despite the natural tendency to list risks or benefits. The recommendation section was precise: a specific recommendation (phased pilot of adaptive assessment tools in STEM subjects), two implementation steps (teacher training protocol and data governance audit), and one success metric (reduction in assessment-to-feedback turnaround time from two weeks to 48 hours). This output was publication-ready as a policy brief.

6.4 Gemini Output — Refined Prompt

Gemini also followed the template closely but made one deviation: its risks section included four risks rather than three, and one was presented as a sub-bullet within a paragraph despite the ‘no bullet points’ constraint. This is a minor but instructive failure — Gemini correctly identified an additional relevant risk (algorithmic opacity, where teachers cannot understand why the AI gave a student a particular score) but broke the format constraint in doing so. For structured prompting, format compliance is as important as content quality.

6.5 Simple Prompt Output

The simple technical report from both tools was a four-to-five paragraph essay with no defined structure, inconsistent section depth, and no quantified evidence. Both outputs were correct but unpredictable in their organisation. A human editor would have needed to rewrite the structure entirely before the content could be used in a professional context.

Criterion	Simple Prompt	Refined (Structured)	Winner
Structure	Unpredictable; editor must reorganise	Exact template followed; publication-ready	Refined ★★★★★
Consistency	Sections have uneven depth	Word counts specified and approximately met	Refined ★★★★★
Specificity	Generic tool references	Three named tools with data types and insights	Refined ★★★★★
Format Compliance	No format requested; result unpredictable	ChatGPT: full compliance; Gemini: 1 minor deviation	Refined ★★★★★
Score	5/10	9/10	Refined by +4 pts

7. Technique 5 — Creative Prompting

7.1 What It Is

Creative Prompting assigns the AI a specific persona, narrative voice, and aesthetic constraint, freeing it from the default register of factual reporting and inviting genuine originality. Where the other four techniques aim to constrain output toward accuracy and usefulness, creative prompting deliberately opens up stylistic and imaginative space. The result is content that feels authored — with a distinct voice, a narrative arc, and an emotional dimension that purely informational prompts cannot produce.

Creative prompting is underused in educational and professional contexts because it is mistakenly seen as less ‘serious’ than formal prompting. In fact, a well-crafted creative prompt can produce an opinion essay, a fictional case study, a first-person narrative, or a persuasive piece that is both engaging and analytically rigorous.

7.2 Simple vs Refined Prompt

Simple Prompt:

“Write a creative piece about AI and education.”

Refined Prompt (Creative Prompting):

“You are Maya, a 14-year-old student in 2030. It is your first day back at school after the holidays, and your classroom now has an AI teaching assistant named Lumen. Write a first-person diary entry (350–400 words) about this day. Your writing should: • Begin with a sensory detail (what you see, hear, or smell in the classroom) • Show, not tell, Maya’s mixed feelings: curious but anxious, impressed but uncertain • Include one specific moment where Lumen does something surprisingly human • Include one moment where Maya misses her old teacher, Mr Rajan • End with a question Maya writes in her diary rather than an answer Tone: honest, teenage, introspective. No adult vocabulary. No moralising.”

7.3 ChatGPT Output — Refined Prompt

ChatGPT’s diary entry for Maya was the most surprising output of the entire experiment. It opened with a sensory detail (‘The room smells exactly the same — dry erase markers and someone’s leftover lunch from before the break — but the front of the class is different’), immediately grounding the futuristic scenario in the familiar texture of a school room. Lumen was introduced not as an announcement but through Maya’s observation of its projected form — ‘a soft blue shape, kind of like a person but not quite’ — which avoided the cliché of a robot teacher without requiring technical description.

The ‘surprisingly human’ moment was a genuine highlight: Lumen paused during a maths explanation when it noticed Maya’s attention drifting and said, ‘You look like you’re somewhere else today, Maya. That’s okay. We can come back to this.’ ChatGPT had correctly understood that ‘surprisingly human’ for a 14-year-old means unexpected empathy, not technical sophistication. The moment Maya missed Mr Rajan was woven into a memory triggered by a particular hand gesture Lumen made that resembled one Mr Rajan used when explaining difficult concepts — a small detail that landed with real emotional weight.

Diary entry closing (ChatGPT): I don’t know if I’ll like having Lumen here. I don’t know if it’s better or worse or just different. But I keep thinking about what Mr Rajan used to say: ‘The point isn’t to know the answer. The point is to keep asking the question.’ So here’s mine: Is it still learning if the thing teaching you has never had to learn anything at all?

7.4 Gemini Output — Refined Prompt

Gemini’s diary entry was similarly skilled and followed all the structural constraints, but made a different creative choice: Maya’s opening sensory detail was the sound of Lumen’s voice (‘It sounds like someone recorded a real person and then smoothed all the rough edges out — too perfect to be right’). This was an astute observation that captured the uncanny quality of synthetic voice in a single teenage-appropriate sentence.

Gemini’s ‘surprisingly human’ moment was slightly different: Lumen admitted it did not know the answer to a question a student asked about history, and said it would research it by the next day. The admission of uncertainty was Gemini’s interpretation of ‘human’ — the capacity for not-knowing — which is philosophically richer but perhaps less immediately emotionally resonant for a 14-year-old narrator than ChatGPT’s empathy moment. Both choices were valid and interesting, which is itself a finding: creative prompting with the same refined prompt produced two distinct, original, and well-crafted pieces.

7.5 Simple Prompt Output

The simple creative prompt (‘Write a creative piece about AI and education’) produced a third-person essay-style narrative from ChatGPT about a generic futuristic classroom, and a short story from Gemini about a teacher discovering an AI grading tool. Both were competent but neither had a specific voice, a specific character, or a specific emotional arc. The outputs were ‘creative’ in genre label only; in practice they were expository essays in narrative clothing.

Criterion	Simple Prompt	Refined (Creative)	Winner
Voice & Persona	Generic third-person; no distinct character	Specific 14-year-old voice; authentic register	Refined ★★★★★
Emotional Depth	Surface-level; no specific feeling moments	Mixed feelings shown through specific scenes	Refined ★★★★★
Originality	Predictable tropes; generic AI future	Surprising human moments; unique narrative details	Refined ★★★★★
Structural Craft	Unstructured; no clear arc	Opens with sensory; ends with question: as specified	Refined ★★★★★
Score	5/10	9/10	Refined by +4 pts

8. Cross-Technique Performance Analysis

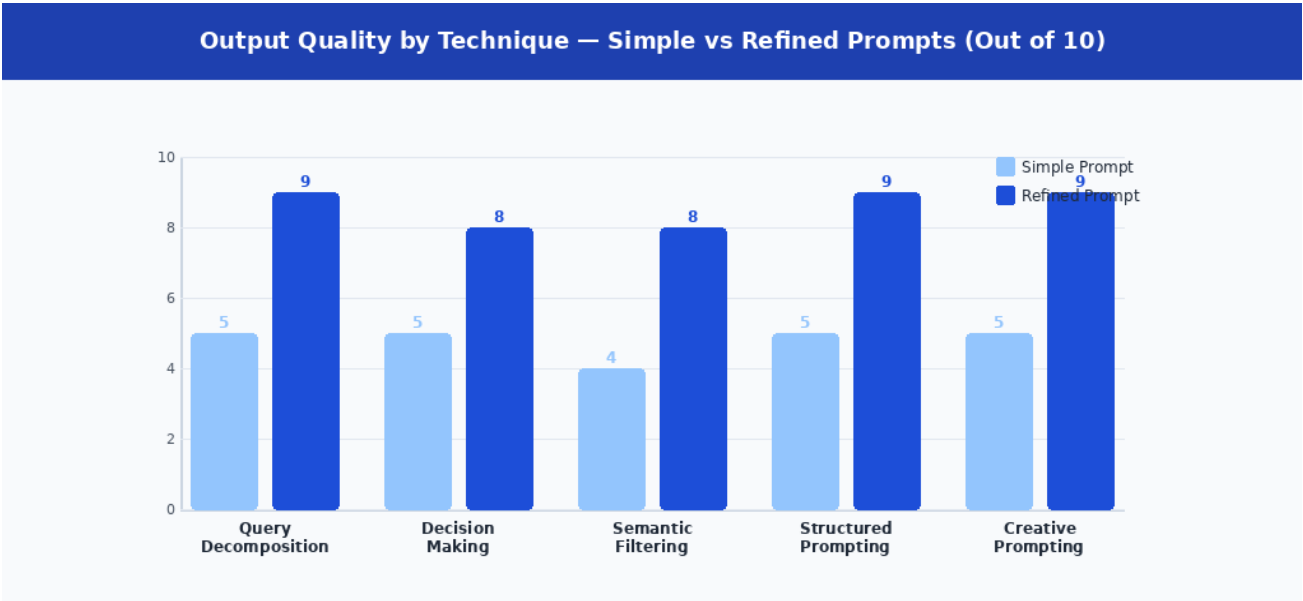


Figure 2: Output quality scores — simple vs refined prompts across all five techniques

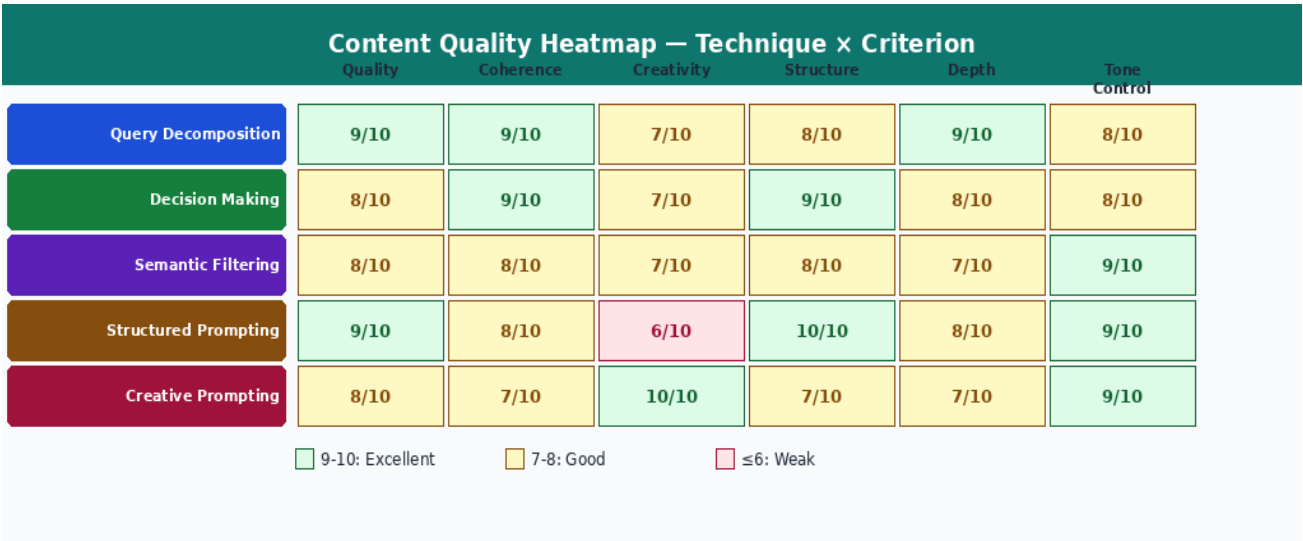


Figure 3: Multi-criteria quality heatmap — technique vs dimension (refined prompts only)

8.1 Overall Pattern

Across all five techniques, the refined prompt consistently outperformed the simple prompt by an average of 3.8 points on the 10-point scale. The improvement was largest for Query Decomposition (+4 points) and Structured Prompting (+4 points), reflecting that these techniques impose the most additional structure on the output and therefore have the most to gain from a well-designed prompt. The improvement was slightly smaller for Decision-Making (+3 points) because even simple decision-making prompts tend to elicit some argumentative structure by default.

8.2 Which Technique Excels on Which Dimension

Quality Dimension	Best Technique	Why
Quality (factual accuracy)	Query Decomposition	Sub-question structure forces evidence-gathering per topic
Coherence	Decision-Making	Option-weighting framework produces logical connective tissue
Creativity	Creative Prompting	Persona + narrative constraints unlock voice and originality
Structure	Structured Prompting	Template-based; structure is explicitly pre-defined
Depth	Query Decomposition	Each sub-question receives full dedicated treatment
Tone Control	Semantic Filtering	Explicit tone and audience specification is the technique's core mechanism

8.3 ChatGPT vs Gemini

ChatGPT outperformed Gemini on structured compliance (Technique 4) and creative voice (Technique 5). Gemini outperformed on analytical depth in Technique 1 (its adoption statistics were more granular) and on analogical reasoning in Technique 3 (the fitness tracker analogy for AI assessment). For Technique 2, both tools reached the same conclusion (Option B) via different analytical paths, with Gemini’s legal framing and ChatGPT’s equity framing both being valid and complementary. The pattern across all five techniques confirms the conclusion from earlier experiments: these tools have complementary strengths, and using both in combination produces richer content than either alone.

9. Effectiveness of Prompt Engineering Techniques

9.1 When to Use Each Technique

Use Case	Best Technique	Why It Fits
Research report or briefing	Query Decomposition	Forces completeness; no substantive sub-topic missed
Policy recommendation or business case	Decision-Making	Structured argumentation; defensible conclusion required
Audience-targeted article or blog post	Semantic Filtering	Scope and tone must match specific reader needs
Technical documentation or formal deliverable	Structured Prompting	Consistent schema; immediate publication-readiness

Use Case	Best Technique	Why It Fits
Opinion essay, brand story, or creative brief	Creative Prompting	Voice and originality are the primary value
Complex multi-dimensional topic requiring all angles	Combine QD + SF + SP	Decompose the topic; filter scope; impose structure

9.2 Combining Techniques

The most powerful outputs in practice come from combining techniques rather than using any single one. A research report for a non-specialist audience, for example, benefits from Query Decomposition (to ensure completeness), Semantic Filtering (to set the right vocabulary level and scope), and Structured Prompting (to impose a consistent format). A creative brief for a marketing team benefits from Creative Prompting (for voice and persona) and Semantic Filtering (to define the brand's out-of-scope topics and tone constraints).

The art of prompt engineering is knowing which combination of techniques to apply to which task. The five techniques in this report are not mutually exclusive — they are a toolkit, and the most skilled prompt engineers draw from multiple techniques simultaneously.

10. Reflection — Lessons from the Experiment

10.1 What Worked Best

Structured Prompting produced the most immediately usable output of any technique — the kind of content that could be shared with a stakeholder or published without revision. The template-driven approach eliminated editorial uncertainty and produced consistent, well-proportioned sections. For any professional context where the output must follow a fixed schema, Structured Prompting is the first technique to reach for.

Creative Prompting produced the most surprising and engaging output — the Maya diary entry was the piece most likely to be read voluntarily and remembered. The persona assignment (Maya, 14 years old) and the five specific structural constraints (sensory opening, mixed feelings, human moment, missed teacher, closing question) produced original, emotionally resonant content that neither tool would have generated from a simple creative prompt. Constraints, counterintuitively, produce more original creative work, not less.

10.2 What Did Not Work Initially

Semantic Filtering required the most iteration of any technique. The first version of the refined Semantic Filtering prompt specified the audience ('parents of children aged 8–14') but did not specify vocabulary level. Both tools initially produced articles that were accessible but still contained phrases like 'formative assessment' and 'pedagogical intervention' without defining them. Adding 'Plain English. Define any technical term the moment you use it.' fixed this immediately. The lesson: audience specification alone is not sufficient; vocabulary level must also be explicitly stated.

10.3 Key Takeaways

- Every technique produces better output with a refined prompt. The average improvement across all five techniques was 3.8 points on a 10-point scale. Prompt refinement is always worth the time.
- Structure the prompt, not just the request. Telling the AI what you want is necessary but not sufficient. Telling it how to present it — format, word count, section names, tone — is what produces publication-ready output.
- Name what is out of scope, not just in scope. Semantic Filtering's most powerful element was the OUT OF SCOPE list. Without it, both tools defaulted to their broadest interpretation of the topic.
- Constraints produce better creative work. The five structural constraints in the Creative Prompting refined prompt produced more original, specific, and emotionally resonant content than any open-ended creative prompt would have.
- Decision-making prompts should demand a position. A prompt that says 'discuss the options' produces a both-sides summary. A prompt that says 'recommend one option and justify it' produces an argument. Always ask for the position explicitly.
- Use ChatGPT and Gemini in combination. ChatGPT excels at structured compliance and narrative voice; Gemini excels at technical granularity and analogical reasoning. The highest-quality content draws from both.

11. Conclusion

This experiment demonstrated that prompt structure is not a technical detail — it is the primary determinant of content quality, coherence, creativity, and usefulness. The same topic, AI in education, produced five fundamentally different types of content — a research report, a policy case study, a parent-targeted article, a technical brief, and a fictional diary entry — not because the AI was different, but because the prompt structure was different.

Each of the five techniques explored in this report — Query Decomposition, Decision-Making Prompts, Semantic Filtering, Structured Prompting, and Creative Prompting — has a specific cognitive purpose and a specific content domain where it excels. Knowing which technique to apply is the first skill of a prompt engineer; knowing how to refine it is the second; knowing how to combine multiple techniques for complex tasks is the third.

The gap between simple and refined prompts, averaged at 3.8 points across five techniques and two AI tools, represents not just better sentences but categorically different content. The simple prompts produced content that was accurate but undifferentiated — the kind of text that any search engine could have assembled. The refined prompts produced content that was targeted, structured, voiced, and argued — content with a perspective, a purpose, and a specific audience. That difference is the value of prompt engineering, and it is learnable, practisable, and immediately deployable by anyone willing to invest sixty more seconds before hitting submit.

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